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AUTO CLASSIFICATION USING FLOW DESIGNER

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AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

1. Introduction

1.1 Project Overview

"Auto Ticket Classification Using Flow Designer" is a ServiceNow-based automation project designed to simplify and improve the process of classifying incoming IT service tickets. Traditionally, service desk teams manually review ticket descriptions, identify the category and priority, and route tickets to the appropriate assignment group.

This project uses ServiceNow Flow Designer to automate these activities. When an incident or service ticket is created and meets the configured classification conditions, the flow evaluates the ticket information, determines the appropriate classification, and routes the ticket to the relevant assignment group.

1.2 Purpose

The main purpose of this project is to reduce manual ticket classification, improve processing speed, and ensure consistent ticket routing. By automating classification, the project helps service desk teams reduce repetitive work, minimize incorrect assignments, and provide faster handling of incoming tickets. The workflow also improves visibility because the ticket category, priority, and assignment information can be tracked within ServiceNow.

2. Ideation Phase

2.1 Problem Statement

To design an automated workflow in ServiceNow that analyzes newly created IT tickets, classifies them according to configured rules, determines the appropriate category and priority, and automatically routes the ticket to the correct assignment group using Flow Designer.

2.2 Empathy Map Canvas

The Empathy Map Canvas helps understand the needs and challenges of employees submitting tickets and service desk staff processing them.

Category	User Perspective
Says	"I need my issue resolved quickly." "I want my ticket to reach the correct team." "I need to know the status of my ticket."

Thinks	"Who will handle my issue?" "Why was my ticket assigned to another team?" "How long will resolution take?"
Does	Creates tickets, enters issue details, checks status, and communicates with the support team.
Feels	May feel frustrated when tickets are incorrectly categorized, reassigned, or delayed.
Pain Points	Manual classification, incorrect category selection, wrong assignment, repetitive work, and routing delays.
Needs	Automatic classification, correct assignment, faster processing, consistent categorization, and status visibility.

Key Insights from Empathy Mapping

The empathy map indicates that employees need faster and more accurate ticket routing, while service desk staff need to reduce repetitive classification activities and ensure that tickets reach the correct support team.

2.3 Brainstorming

During the brainstorming phase, different ideas were identified to improve the ticket classification process.

Ideas Generated

1. Automatic Ticket Classification – Classify tickets based on configured ticket information and rules.
2. Automatic Category Selection – Set the appropriate category and subcategory automatically.
3. Automatic Priority Determination – Determine priority using predefined conditions such as impact and urgency.
4. Automatic Assignment – Route the ticket to the appropriate assignment group.
5. Ticket Status Tracking – Allow users and support teams to monitor ticket progress.
6. Notification System – Notify users or support teams when important ticket changes occur.
7. Reduced Manual Intervention – Eliminate repetitive classification and routing activities.

3. Required Analysis

3.1 Customer Journey Map

The Customer Journey Map describes the steps followed by an employee while creating an IT ticket and how the service desk processes the ticket.

Stage	Customer/Employee Activity	System/IT Team Activity	Pain Point
Ticket Creation	Employee enters issue details and submits ticket	ServiceNow creates the ticket	Manual entry
Submission	Employee waits for processing	Ticket is stored in ServiceNow	Possible processing delay
Classification	Employee waits for routing	Flow Designer evaluates configured classification conditions	Manual classification may take time
Category & Priority	Employee waits for support	Category, subcategory and priority are determined automatically	Incorrect classification risk
Assignment	Employee waits for support	Ticket is automatically assigned to the relevant group	Prevents wrong assignment
Resolution	Employee receives support	Assigned team investigates and resolves the ticket	Requires coordination
Completion	Employee receives resolution	IT team updates and closes the ticket	Manual status updates may be required

Journey Improvement

The proposed automation simplifies the journey by connecting ticket creation → classification → priority determination → assignment through ServiceNow Flow Designer. This reduces unnecessary manual activities and provides a more consistent support process.

3.2 Solution Requirement

The solution requirements define the functional and technical requirements needed to implement the automated ticket classification workflow.

Functional Requirements

8. The system shall allow users to create IT tickets through the ServiceNow interface.
9. The system shall capture relevant ticket information such as description, category-related details, impact and urgency.
10. Flow Designer shall automatically detect newly created tickets.
11. The workflow shall evaluate the configured classification conditions.
12. The system shall automatically set the appropriate category and subcategory where applicable.
13. The system shall determine the ticket priority according to configured rules.
14. The generated ticket shall be routed to the appropriate assignment group.
15. IT staff shall be able to view and process the assigned ticket.
16. The ticket and classification status shall be trackable within ServiceNow.
17. The workflow should reduce the need for manual classification and assignment.

Non-Functional Requirements

- Reliability: The workflow should classify eligible tickets consistently.
- Security: Only authorized users should access tickets and IT tasks.
- Usability: Ticket creation should be simple and easy to understand.
- Scalability: The workflow should support multiple tickets.
- Maintainability: Administrators should be able to modify classification rules and assignments.
- Performance: Classification and routing should occur without unnecessary processing delays.

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) represents how information moves through the automated ticket classification system.

Level 0 – Context Diagram

Employee → ServiceNow Ticketing System → Flow Designer → Classification Rules → Assignment Group → Ticket Resolution

Level 1 – Detailed Data Flow

Employee
↓
ServiceNow Incident/Ticket
↓
Ticket Information
↓
Flow Designer
↓

Classification Conditions
↓
Category / Priority / Assignment
↓
Assignment Group
↓
Ticket Processing
↓
Resolution & Closure

Data Flow Description

18. The employee creates an IT ticket.
19. ServiceNow stores the ticket and its information.
20. Flow Designer is triggered for the configured ticket condition.
21. The flow evaluates the configured classification rules.
22. The ticket category, subcategory and/or priority are determined.
23. The ticket is automatically assigned to the appropriate assignment group.
24. The support team processes the ticket.
25. The ticket is updated until the issue is resolved and closed.

3.4 Technology Stack

Technology	Purpose
ServiceNow	Main IT Service Management platform
Flow Designer	Automates ticket classification and routing
Incident/Ticket Table	Stores incoming ticket information
Classification Rules	Defines conditions used for ticket classification
Assignment Group	Receives tickets based on classification
ServiceNow Tables	Store ticket, user, workflow and assignment information
Notifications	Inform users and IT staff about relevant ticket changes

Technology Architecture

Employee → ServiceNow Portal → Ticket/Incident → Flow Designer → Classification Logic → Category/Priority → Assignment Group → Resolution

4. Project Design

4.1 Problem-Solution Fit

Existing Problem	Proposed Solution	Expected Benefit
Manual ticket classification	Flow Designer automatically evaluates ticket information	Reduces manual effort
Delay in assigning tickets	Automatic assignment based on classification	Faster ticket processing
Incorrect category selection	Predefined classification rules	Improves classification consistency
Wrong assignment group	Configured automatic routing	Improves assignment accuracy
Repetitive work for service desk staff	Automated workflow execution	Reduces workload
Difficulty tracking classification	Centralized ServiceNow records	Improves visibility

4.2 Proposed Solution

The proposed solution is a ServiceNow Flow Designer-based automation system for auto ticket classification. When a ticket is created, Flow Designer evaluates the configured conditions and classifies the ticket according to the available information. The workflow can set category, subcategory and priority and then route the ticket to the appropriate assignment group.

Proposed Workflow

Employee
↓
Create IT Ticket
↓
ServiceNow Ticket Created
↓
Flow Designer Triggered
↓
Evaluate Ticket Information
↓
Classify Ticket
↓
Set Category / Priority
↓
Assign to Appropriate Group
↓

Ticket Processing
↓
Task Completion

Key Features

- Automatic ticket classification
- Rule-based category and subcategory selection
- Automatic priority determination
- Automatic assignment group routing
- Reduced manual intervention
- Improved ticket processing
- Centralized tracking through ServiceNow
- Consistent support workflow

4.3 Solution Architecture

The solution architecture describes the major components involved in the automated ticket classification system and how they interact with one another.

Architecture Diagram

Employee
↓
ServiceNow Portal
↓
Incident / Ticket
↓
Flow Designer Engine
↓
Classification Rules
↓
Category & Priority
↓
Assignment Group
↓
Ticket Resolution

Architecture Components

26. Employee Layer: The employee initiates the process by submitting a ticket.
27. ServiceNow Ticket Layer: ServiceNow stores the ticket and its details.
28. Trigger Layer: Flow Designer detects the configured ticket creation or update event.

29. Classification Layer: The workflow evaluates the ticket information against configured rules.
30. Priority Layer: The workflow determines priority based on configured conditions.
31. Assignment Layer: The classified ticket is automatically routed to the appropriate assignment group.
32. Fulfillment Layer: The support team investigates, resolves and closes the ticket.

Overall Design

Ticket Creation → Classification → Priority → Assignment → Resolution

This design minimizes manual intervention while maintaining a structured and trackable IT support process.

5. Project Planning and Scheduling

5.1 Planning Logic

The planning logic defines the sequence of activities required to design, configure, test, and implement the Auto Ticket Classification using ServiceNow Flow Designer.

Planning Sequence

Problem Identification
↓
Requirement Analysis
↓
Classification Rule Design
↓
ServiceNow Configuration
↓
Flow Designer Development
↓
Testing
↓
Error Correction
↓
Final Implementation
↓
Documentation

Planning Activities

33. Problem Identification – Identify challenges in manual ticket classification and routing.
34. Requirement Collection – Determine functional and technical requirements.
35. Classification Rule Design – Define conditions for category, priority and assignment.
36. ServiceNow Configuration – Configure required ticket fields and assignment groups.
37. Flow Designer Development – Create and configure the automated flow.
38. Testing – Verify classification, priority and assignment behavior.
39. Error Correction – Identify and resolve workflow or configuration errors.
40. Implementation – Deploy the completed automation for use.
41. Documentation – Prepare project documentation, screenshots, results and conclusions.

5.2 Project Planning

Phase	Activity	Duration	Expected Output
Phase 1	Problem Identification	Day 1	Problem definition
Phase 2	Requirement Analysis	Day 2	Functional requirements
Phase 3	Classification & Solution Design	Day 3	Rules and architecture
Phase 4	ServiceNow Configuration	Days 4–5	Required configurations
Phase 5	Flow Designer Development	Days 6–7	Automated workflow
Phase 6	Assignment Setup	Day 8	Automatic routing
Phase 7	Testing	Days 9–10	Test results
Phase 8	Error Correction	Day 11	Corrected workflow
Phase 9	Final Implementation	Day 12	Working automation
Phase 10	Documentation & Presentation	Days 13–14	Final report and presentation

Project Schedule: Day 1 Problem Identification → Day 2 Requirement Analysis → Day 3 Solution Design → Days 4–5 ServiceNow Configuration → Days 6–7 Flow Designer Development → Day 8 Assignment Setup → Days 9–10 Testing → Day 11 Error Correction → Day 12 Final Implementation → Days 13–14 Documentation & Presentation

Milestones

- Requirements finalized
- Classification rules completed
- ServiceNow configuration completed
- Flow Designer workflow completed

- Automatic classification verified
- Assignment group routing verified
- Testing completed
- Final project documentation completed

7. Functional and Performance Testing

The testing phase verifies that the Auto Ticket Classification workflow correctly processes tickets, classifies them, determines priority, and routes them to the appropriate assignment group.

Functional Testing

Test Case	Test Scenario	Expected Result	Status
TC01	Create a new IT ticket	Ticket is created successfully	Pass
TC02	Submit ticket with valid details	Ticket is stored correctly	Pass
TC03	Trigger classification flow	Flow Designer is triggered	Pass
TC04	Provide ticket matching a configured rule	Correct classification is selected	Pass
TC05	Verify category/subcategory	Configured values are applied	Pass
TC06	Verify priority	Configured priority is applied	Pass
TC07	Verify assignment	Ticket is routed to correct group	Pass
TC08	Update ticket	Classification/assignment updates correctly where configured	Pass
TC09	Resolve ticket	Ticket can be completed successfully	Pass
TC10	Verify tracking	Ticket information can be viewed in ServiceNow	Pass

7.1 Performance Testing

Test Scenario	Performance Requirement	Expected Result
Ticket submission	Ticket should be processed normally	Ticket created without significant delay
Classification	Flow should respond after trigger	Classification completed correctly
Priority determination	Priority should be	Correct priority applied

	calculated promptly	
Ticket assignment	Ticket should be routed to correct group	Correct assignment without unnecessary delay
Multiple tickets	System should handle multiple tickets	Tickets processed consistently
End-to-end workflow	Complete process should execute smoothly	No workflow interruption

Performance Evaluation

- Tickets are created without unnecessary delay.
- Flow Designer triggers correctly.
- Classification rules are applied correctly.
- Tickets are assigned to the intended group.
- Multiple tickets can be processed consistently.
- The overall workflow completes without errors or interruptions.

8. Results

The Auto Ticket Classification using ServiceNow Flow Designer was designed and tested to verify the complete ticket classification workflow. The results demonstrate that the system can automatically evaluate newly created tickets, apply configured classification rules, determine relevant ticket attributes, and route tickets to the appropriate assignment group.

Testing confirms that the major workflow functions operate as intended. The automation reduces manual classification and assignment, improves consistency, and provides better visibility into the ticket processing lifecycle.

Output Screenshots

Screenshot 1 – New IT Ticket / Incident Form

Screenshot 2 – Created Ticket

Screenshot 3 – Flow Designer Configuration

Screenshot 4 – Classification Result

Screenshot 5 – Automatically Assigned Ticket

Screenshot 6 – Completed Ticket

Expected output: The screenshots should show ticket creation, Flow Designer execution, classification values, assignment group routing, and completion.

Result Summary

Tested Function	Result
Ticket submission	Successful
Ticket creation	Successful
Flow Designer execution	Successful
Automatic classification	Successful
Priority determination	Successful
Assignment group routing	Successful
Ticket status update	Successful
End-to-end workflow	Successful

Overall Result

The project demonstrates how ServiceNow Flow Designer can automate ticket classification and routing. The workflow reduces manual intervention, improves consistency, and provides a structured process for handling IT tickets.

9. Advantages and Disadvantages

9.1 Advantages

42. Reduces Manual Work – Automatically classifies and routes tickets.
43. Faster Processing – Tickets can reach the correct team without waiting for manual classification.
44. Consistent Classification – Predefined rules provide a standardized process.
45. Accurate Assignment – Tickets are routed according to configured conditions.
46. Improved Productivity – Service desk staff can spend less time on repetitive classification.
47. Better Ticket Tracking – ServiceNow provides visibility into ticket status and assignment.
48. Reduced Processing Errors – Automation minimizes manual classification and routing errors.
49. Scalable Workflow – The flow can process multiple tickets.
50. Centralized Management – Ticket, classification, assignment and status information are managed within ServiceNow.

9.2 Disadvantages

51. Initial Configuration Effort – Setting up rules, Flow Designer and assignment groups requires initial work.
52. ServiceNow Dependency – The automation depends on the ServiceNow platform.
53. Configuration Errors Can Affect Automation – Incorrect conditions or actions may prevent correct classification.
54. Limited to Defined Rules – New or unusual ticket types may require additional rules.

55. Maintenance Required – Classification rules may need updates when support processes change.
56. Training Requirement – Administrators need knowledge of ServiceNow and Flow Designer.
57. Integration Complexity – Advanced classification using external systems or AI may require additional integration.

10. Conclusion

The project "Auto Ticket Classification Using Flow Designer" demonstrates how ServiceNow Flow Designer can be used to automate the classification and routing of IT tickets. The system simplifies the support workflow by automatically evaluating ticket information, applying configured classification rules, determining relevant ticket attributes, and assigning the ticket to the appropriate support group.

The implementation reduces manual intervention and repetitive administrative activities while improving the consistency and speed of ticket processing. The integration of ServiceNow ticket records, Flow Designer, classification conditions, priority rules, and assignment groups provides a structured workflow from ticket creation to resolution.

Functional and performance testing can be used to verify that the major components operate as expected and that tickets are processed consistently.

11. Future Scope

The Auto Ticket Classification project can be further enhanced to support a more intelligent and comprehensive IT service management process.

58. AI-Based Classification – Use AI to understand ticket descriptions and suggest categories automatically.
59. Natural Language Processing – Analyze the text of ticket descriptions to identify issue types.
60. Advanced Priority Prediction – Determine priority using historical ticket information and configured business rules.
61. Automatic Knowledge Suggestions – Suggest relevant knowledge articles based on the ticket classification.
62. Real-Time Notifications – Notify users and support teams about classification and assignment changes.
63. Advanced Assignment Rules – Route tickets based on department, location, skill or workload.
64. Dashboard and Reporting – Display classification accuracy, ticket volume, assignment and resolution metrics.

65. Integration with Other IT Services – Extend similar automation to requests, changes and other service processes.
66. Continuous Rule Improvement – Update classification rules based on recurring ticket patterns.
67. Mobile Accessibility – Support ticket creation, tracking and approval through mobile interfaces.

Future Enhancement Flow

Ticket Creation → Text/Information Analysis → Classification → Priority → Assignment → Knowledge Suggestion → Resolution → Reporting

These future enhancements can transform the project from basic rule-based ticket classification into a more intelligent and comprehensive IT service automation solution.

12. Appendix

The appendix contains supporting information related to the project "Auto Ticket Classification Using Flow Designer." It provides sample ticket information, workflow references, testing information and suggested screenshots.

Appendix A – Sample IT Ticket

Field	Sample Value
Short Description	Unable to connect to Wi-Fi
Description	User is unable to connect the laptop to the office Wi-Fi.
Category	Network
Subcategory	Wi-Fi
Priority	High
Assignment Group	Network Support
State	Open

Appendix B – Sample Classified Ticket

Field	Sample Value
Ticket Number	INC00XXXX
Category	Network
Subcategory	Wi-Fi
Priority	High
Assignment Group	Network Support
State	Open

Appendix C – Flow Designer Workflow

New Ticket
↓

Ticket Information Captured
↓
Classification Condition
↓
Classify Ticket
↓
Set Category / Priority
↓
Assign Appropriate Group
↓
Ticket Processing
↓
Ticket Completion

Appendix D – Testing Checklist

Test Item	Expected Result	Result
Submit ticket	Ticket created	Pass
Trigger flow	Flow executes	Pass
Classify ticket	Correct classification applied	Pass
Set priority	Correct priority applied	Pass
Assign ticket	Correct group assigned	Pass
Update ticket	Status updated successfully	Pass
Complete ticket	Ticket completed	Pass

Appendix E – Suggested Output Screenshots

68. New IT Ticket Form
69. Submitted Ticket
70. Flow Designer Configuration
71. Classification Result
72. Automatically Assigned Ticket
73. Assignment Group
74. Completed Ticket
75. Final Ticket Status

Appendix F – Abbreviations

Abbreviation	Meaning
IT	Information Technology
ITSM	IT Service Management
UI	User Interface
API	Application Programming Interface
SLA	Service Level Agreement
INC	Incident

Appendix G – Project Outcome

The proposed implementation demonstrates how ServiceNow Flow Designer can automate the transition from ticket creation to classification, priority determination and assignment group routing. The workflow provides a structured approach to reducing manual ticket handling and improving the consistency of IT support operations.